Strategic Menu Planning and Cost Optimization for Juice Bar Success

FINAL REPORT

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Section 1. Executive Summary

This project was conducted to enhance product offerings, analyze customer behavior, and inform operational decisions for a local food and beverage establishment by utilizing sales and cost data gathered over a nine-month period. The methodology encompassed structured data preparation, SKU categorization, and the application of analytical techniques, including Pareto analysis, time-series visualization, and gross-profit-margin evaluation.

The initial phase of analysis concentrated on data cleaning and standardization of SKU codes, with each SKU systematically decoded to indicate its category (food or drink), subcategory (e.g., Sandwich, Juice), and unique identifier. Sales data was segmented into 18 half-monthly intervals to accommodate the seasonal rotation of menu items. A Pareto chart analysis was subsequently employed to classify SKUs into high, medium, and low performers based on normalized revenue and unit sales, adjusted for varying product availability across intervals.

The results yielded key insights, revealing that approximately 38% of SKUs, such as DM06 (Milk Soda), DJ03 (Mix Juice), and FP01 (Aloo Patty), account for nearly 80% of the revenue and unit sales, highlighting their significance as core offerings. Additionally, revenue trends exhibited a stronger correlation with SKU variety than with the number of working days, underscoring the importance of product assortment over operating hours.

Customer behavior analysis indicated that products priced at ₹40 achieved the highest normalized unit sales, suggesting a psychological sweet spot. Food–drink pairing patterns revealed strong preferences for sandwich–shake combinations, particularly those involving Paneer Corn or Veg Sandwich with Oreo or Custom shakes. Time-of-day and weekday analysis identified the 4PM–10PM window and Monday–Tuesday slots as critical for sales and inventory restocking, while extended holiday closures negatively affected post-holiday revenue.

Variable costs, such as disposables and ice, were found to be seasonally influenced and candidates for improved management. The project concludes with actionable recommendations to retain top-performing SKUs, bundle slow-moving but high-revenue items, optimize pricing, restructure restocking schedules, and reduce unnecessary costs. These strategies, grounded in empirical data and visual analytics, provide a robust foundation for enhancing profitability and operational efficiency in small-scale food businesses.

Section 2. Detailed Explanation of Analysis Methods

2.1 Data Collection & Preparation

The quality of the analysis and insights depends largely on the quality of the data. The first steps were to cleanup and pre-process the data in the required formats and to ensure that all data is complete and have consistent units. After this, the data was codified based on the SKUs for better interpretation of the reports.

To interpret an SKU code, start by examining the first character, which indicates whether the item is a Food (F) or a Drink (D). The first two letters define the subcategory, such as FP for FoodPatty, FB for FoodBurger, FS for FoodSandwich, FH for Food Healthy, DF for DrinkFruits, DC for DrinkCoffee & Confectionery, DS for DrinkSyrups, DM for DrinkMocktails, and DJ for DrinkJuice. The numerical suffix, like 01 in FH01, serves as a unique identifier for the specific product within its category. For example, FH01 is a Food product in the FoodHealthy category and is the first unique item listed under this category

Having prepared the data, the analysis steps for each problem statement were finalized as follows.

2.2 Optimize menu items by categorizing products into low selling and high selling based on sales throughout all the seasons.

Analyze Menu Items: In this initial step, the products (SKUs) that are available for sale in the menu are identified. Since the products are seasonal, they change every 2.5/2/1.5 months depending on the situation. So, using highest common factor method, sales data was binned into half a month intervals because in this interval we are confident that menu items will not change and compare these intervals using revenue, units sales to perform analysis.

Pareto chart analysis for SKU classification: Starting with normalizing the revenue of each SKU with number of intervals it sold because some products are seasonable and some were introduced in the middle stage of analysis and hence contribute less in terms of time intervals they sold compare to other products that were generating from first day of starting of the shop.
Based on normalized revenue and normalized unit sales, Pareto chart for both these factor for each SKU will be created by excluding some outliers that may start in the end of last interval like 2-3 days or just 1 interval because it will not show clear picture to reduce them or retain them.

Classification of products based on revenue/unit sales contribution as follows:

- High revenue/units sold products: Contributing more than or equal to 80% in Pareto chart
- Medium revenue/units sold products: Contributing between 20 % and 80% in Pareto chart
- Low revenue/units sold products: Contributing more than or equal to 20% in Pareto chart

Units Sale\ Revenue	High Revenue	Medium Revenue	Low Revenue
High Unit Sales	Must be kept with different varieties	Need to change price	can be use in combo with low sale items
Medium Unit Sales	Need to change advertise	Must contain	Focus on increasing price
Low Unit Sales	Offers to increase units sales	Must be change or upgrade	Should be removed unless started yet

Analyzing revenue in each intervals: To understand how revenue is influenced by operational factors (working days) and menu items moved (unique SKU), a facet chart was used to visualize Revenue, Working Day, Unique products sold across bi-monthly intervals. These normalized metrics provide insights into the efficiency and drivers of revenue generation.

2.3 Analyzing customer behavior in terms of prices, off days and food & drink combinations

Binned products based on Price: To analyze the relationship between product prices and sales performance, the approach involves binning SKUs based on price. For each SKU, calculate the "Normalized Unit Sale," which is the total units sold divided by the number of bi-monthly intervals the product was available. Group SKUs with the same price into bins. Visualize the data using a combination of a bar graph and a line chart. The bar graph will represent the total sum of normalized unit sales for each price bin, while the line chart will show the number of products within each bin. This method allows for a clear comparison of sales performance across different price categories.

Correlation between Food and Drinks: To make our sales shine during off-peak hours, we're putting together a super cool plan. We'll start by asking our customers what they love to eat and drink together. Then, we'll use a special heat map to figure out what they order the most. With all this info, we'll create a menu that's packed with delicious pairings. This way, we can make sure our customers are happy and our sales go up during those slower times.

Sales patterns based on holiday streaks were analyzed using two methods: box plots and bar graphs. The box plot examined the impact of consecutive holidays—1, 2, and 3—on unit sales up to the next holiday. This visual representation provided insights into the distribution and variability of sales during these periods. Additionally, a bar graph was used to assess the sales trends before and after each holiday streak, specifically looking at the percentage change in sales three days prior and three days post-holiday. This method helped identify the magnitude of sales fluctuations during these holiday periods and its impact on customer retention with the shop.

Understanding Customer behavior based on time of the day and day of the week: Now, let's use time series analysis to figure out the best time to buy inventory for a shop based on day of week and time of . We'll use box plots for each day of the week to see how the average unit sales are distributed. Then, we'll figure out which day is the best day to buy inventory, except for Sunday when some shops are closed for buying raw food items. Then using line chart to figure out sales based on morning preference of customers or evening preferences to do find out best time of the day to purchase inventory.

2.4 Understanding Costs for food and drinks and fixed and variable operational cost

Understanding Goods Cost: Let's break down the cost of food and drinks to understand the numbers better. First, let's group the products based on what they have in common. For drinks, we have milk drinks, juices, and lemon drinks. For food, we have patties, sandwiches, and sweet corns.

Understanding Gross Profit Margin: Next, let's look at the Gross Profit Margin (GPM) for each category throughout the season. GPM is calculated by dividing the Gross Profit by the Net Sales and multiplying by 100. This helps us see which categories have high GPM and high unit sales. By thinking about the opportunity cost, we can choose the right products that make the most money.

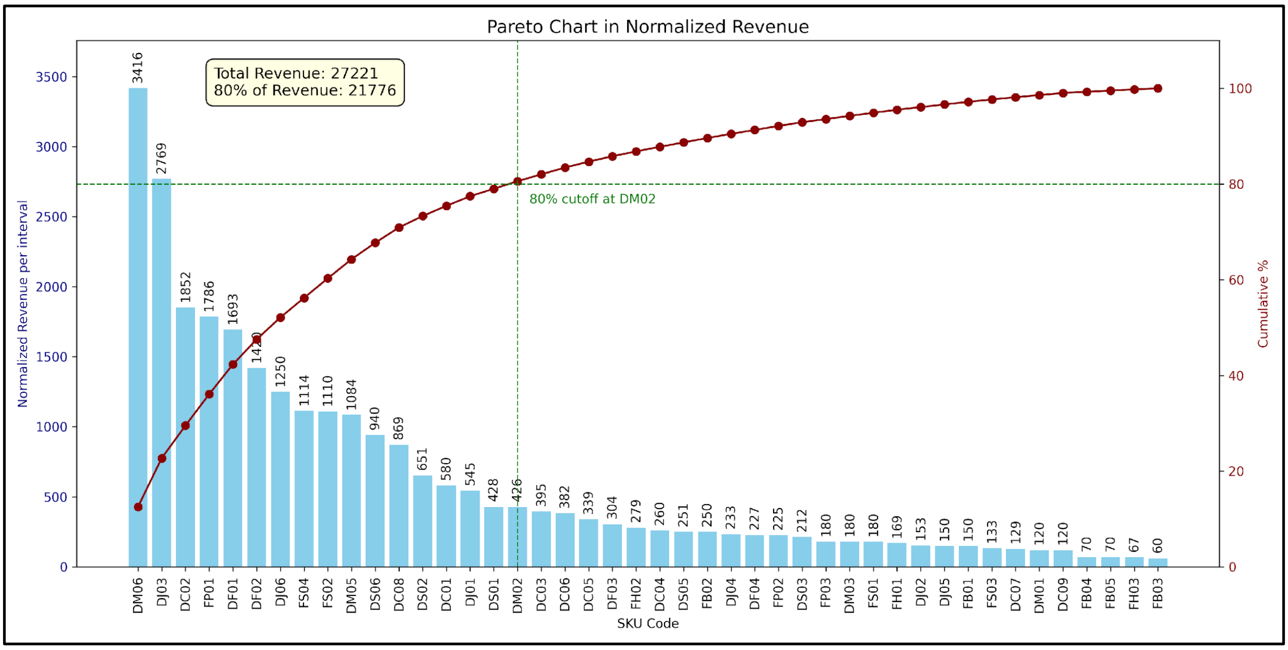
Fixed Cost Analysis: By finding out time for fixed cost and to have a good cash flow, we need to look at how much to spend and in what time period like month wise, season wise depending on weather and sales trend corresponding to the season and whether to invest in shops for further improvement by looking at sales units that go above threshold and predicting whether to invest on some product or not.

Variable Cost Analysis: Key components like electricity, gas cylinders, fuel, ice, and disposables fluctuate monthly and are influenced by seasonal sales. Examining the relationship between electricity and gas usage helps determine the most cost-effective food preparation method, optimizing resource allocation for maximum ROI. Analyzing market visit frequency and necessity reduces fuel and travel expenses, optimizing supply chain logistics. Aligning ice stock levels with seasonal demand patterns better manages ice expenditure, closely tied to beverage sales. This targeted analysis supports profitability through data-driven strategic cost adjustments.

Section 3.Results and finding

3.1 Analyzing Revenue distribution in half a month interval across 9 months to identify the products most demanded in all seasons

Fig.1 Pareto Chart for Normalized Revenue

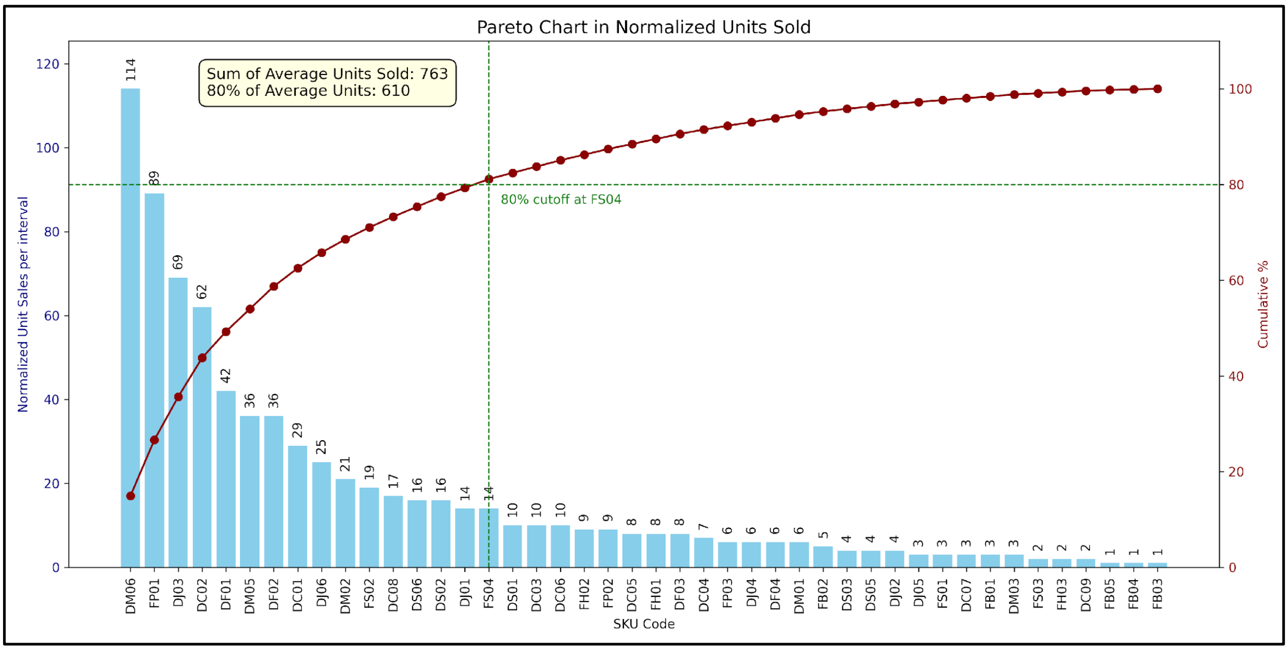


The normalized revenue distribution across 18 half-monthly intervals reveals that approximately **38% of the** SKUs contribute to about **80%** of the total revenue. These high-revenue SKUs include DM06 (Milk Soda), DJ03 (Mix Juice), DC02 (Hot Coffee M), FP01 (Aloo Patty), DF01 (Banana Shake), DF02 (Mango Shake), DJ06 (Mousambi Juice), FS04 (Paneer Corn Sandwich), FS02 (Veg Sandwich), and DM05 (Lemon Soda).

The remaining revenue is generated by SKUs in the medium revenue contribution category, which spans from **20% to 80%** of the total revenue. This group includes DS06 (Chocolate Shake), DC08 (Oreo Shake), DS02 (Strawberry Shake), DC01 (Hot Coffee S), DJ01 (Carrot Juice), DS01 (Butterscotch Shake), DM02 (Lemonade), DC03 (Cold Coffee), DC06 (Chocolate Coffee), DC05 (Butterscotch Coffee), DF03 (Papaya Shake), FH02 (M Sweet Corns), DC04 (Custom Coffee), DS05 (Blueberry Shake), FB02 (Veg Burger), DJ04 (Pineapple Juice), DF04 (Custom Shake), FP02 (Noodle Patty), DS03 (Black current Shake), FP03 (Paneer Patty), DM03 (Virgin Mojito), FS01 (Aloo Tikki Sandwich), FH01 (S Sweet Corns), DJ02 (Kinu Juice), DJ05 (Apple Juice), FB01 (Cream Burger), FS03 (Paneer Onion Sandwich), DC07 (Hot Coffee L), DM01 (1L Water Bottle), and DC09 (Kit kat Shake).

The **low revenue** contribution category, comprising the **bottom 20%** of SKUs, includes FB04 (Paneer Corn Burger), FB05 (Paneer Onion Burger), FH03 (L Sweet Corns), and FB03 (Double Tikki Burger).

Figure 2 - Pareto chart for Normalized Units Sales



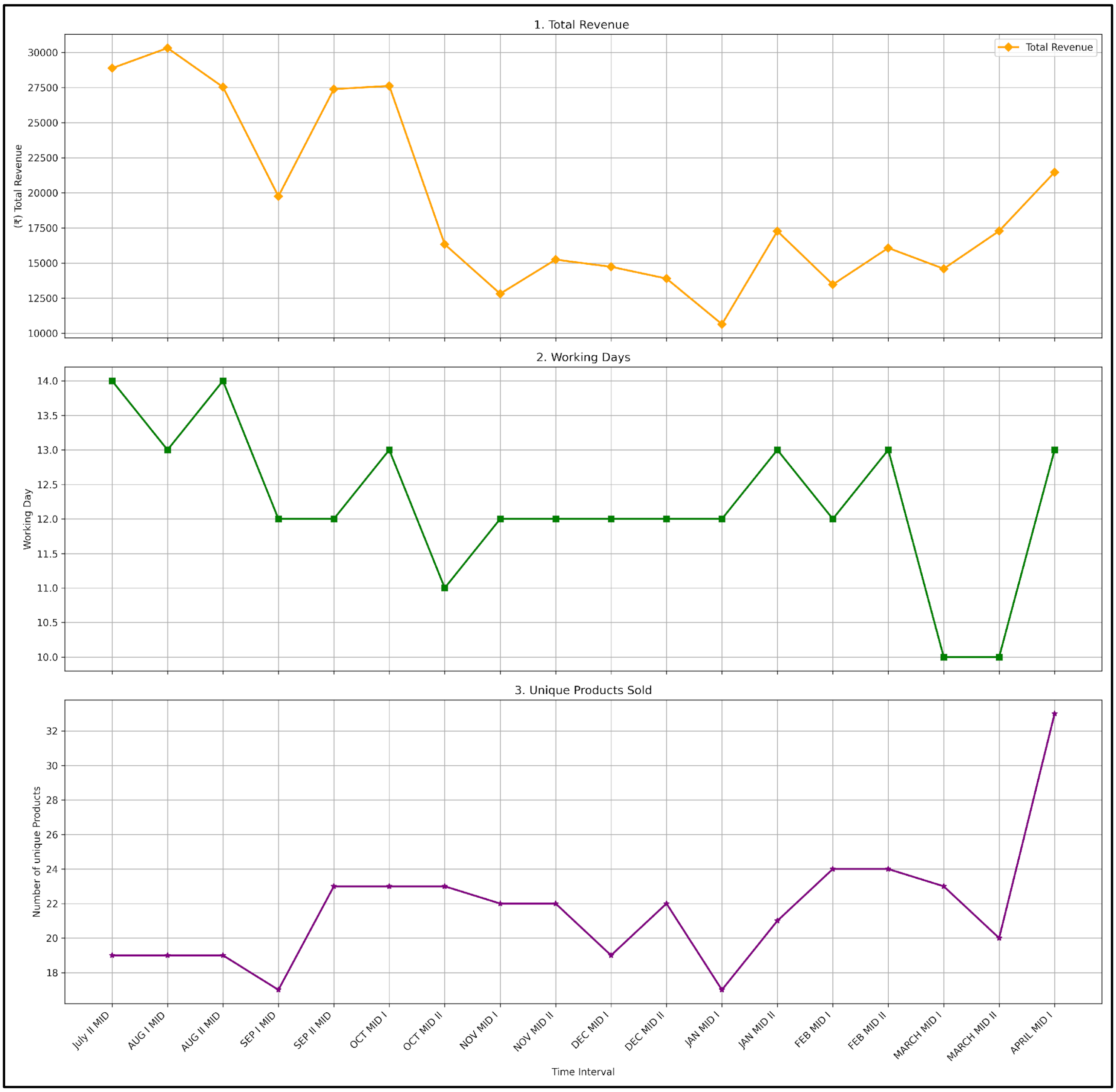
The Pareto analysis conducted over 18 half-monthly intervals categorizes products based on their contribution to the total units sold.

The **top 35%** of products, collectively accounting for **approximately 80%** of the total units sold, comprise DM06 (Milk Soda), FP01 (Aloo Patty), DJ03 (Mix Juice), DF01 (Banana Shake), DC02 (Hot Coffee M), DM05 (Lemon Soda), DC08 (Oreo Shake), DS02 (Strawberry Shake), DS06 (Chocolate Shake), FS02 (Veg Sandwich), DJ06 (Mousambi Juice), DM02 (Lemonade), FS04 (Paneer Corn Sandwich), DC06 (Chocolate Coffee), DC03 (Cold Coffee), DS01 (Butterscotch Shake), DF02 (Mango Shake), and DF03 (Papaya Shake).

The **medium** contribution category, encompassing **20% to 80%** of the units sold, includes DC05 (Butterscotch Coffee), DJ01 (Carrot Juice), DF04 (Custom Shake), FH02 (M Sweet Corns), FH01 (S Sweet Corns), DS03 (Black current Shake), DS05 (Blueberry Shake), DJ04 (Pineapple Juice), DC01 (Hot Coffee S), DJ02 (Kinu Juice), DJ05 (Apple Juice), DC07 (Hot Coffee L), DM01 (1L Water Bottle), DC04 (Custom Coffee), FP03 (Paneer Patty), FS01 (Aloo Tikki Sandwich), FP02 (Noodle Patty), FS03 (Paneer Onion Sandwich), FB02 (Veg Burger), FH03 (L Sweet Corns), DM03 (Virgin Mojito), FB01 (Cream Burger), and DC09 (Kit kat Shake).

Finally, the **low contribution** category, representing **less than 20%** of the units sold, comprises FB05 (Paneer Onion Burger), FB04 (Paneer Corn Burger), and FB03 (Double Tikki Burger).

Fig.3 Facet chart for Revenue drivability with products and working days

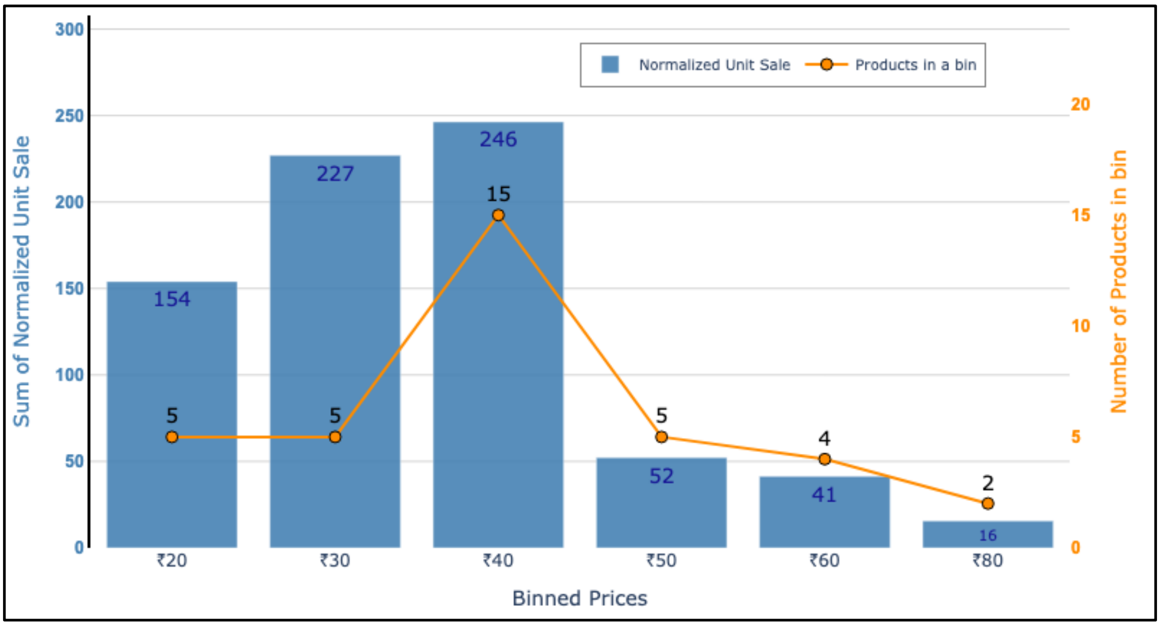


Total Revenue vs. Working Days: Revenue does not increase proportionally with working days. Even when intervals have similar or higher working days, revenue can drop significantly (e.g., intervals in Nov 'H1' and Dec 'H1' have 12 working days but lower revenues compared to earlier periods with similar or fewer working days). This suggests that just increasing the number of working days does not guarantee higher revenue.

Total Revenue vs. Unique Products Sold: There is a more direct relationship between the number of unique products sold and total revenue. In intervals where the number of unique products spikes (such as Apr 'H2'), there is a noticeable increase in revenue, even if working days remain the same or are lower. Similarly, periods with fewer unique products tend to correspond with lower revenue, despite having the same or more working days.

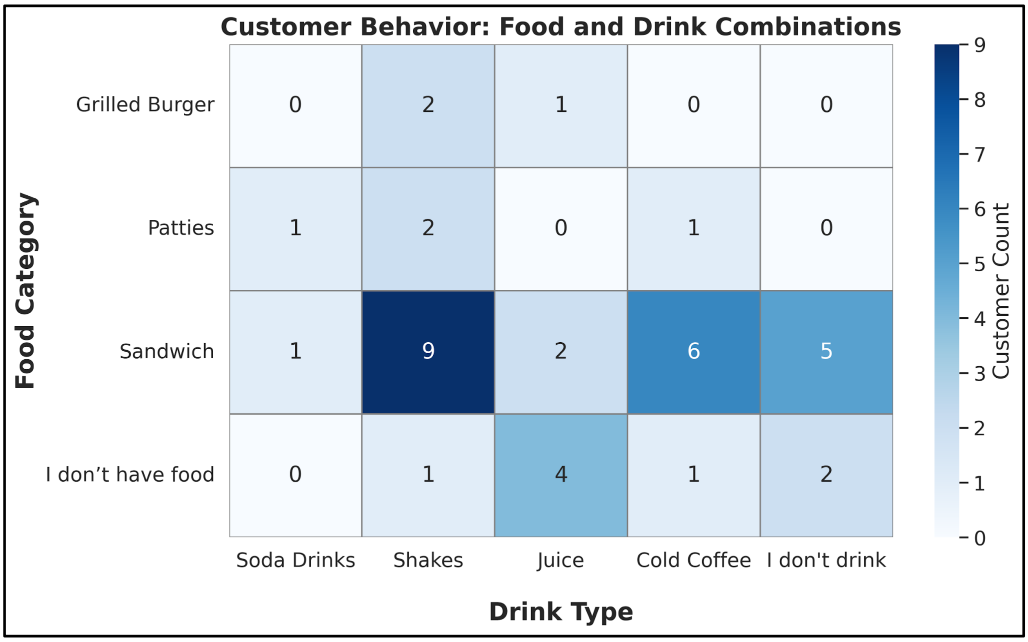
3.2 Analyzing customer behavior while purchasing products

Fig.4 Normalized Sales of all products grouped by price



The results indicate that products priced at ₹40 exhibit the highest Normalized Unit Sales, outperforming other price categories. This price point also hosts the largest number of unique products, with 15 options, which may enhance its appeal and contribute to its strong sales performance. Additionally, the ₹20 and ₹30 price bins show significant customer interest, with relatively high Normalized Unit Sales and a moderate selection of 5 products each. **Conversely**, as prices rise above ₹40, both the Normalized Unit Sales and the number of available products tend to decline. Notably, the ₹80 price bin records very low normalized sales and offers only 2 products, suggesting that higher-priced items are less favored or have a limited market presence.

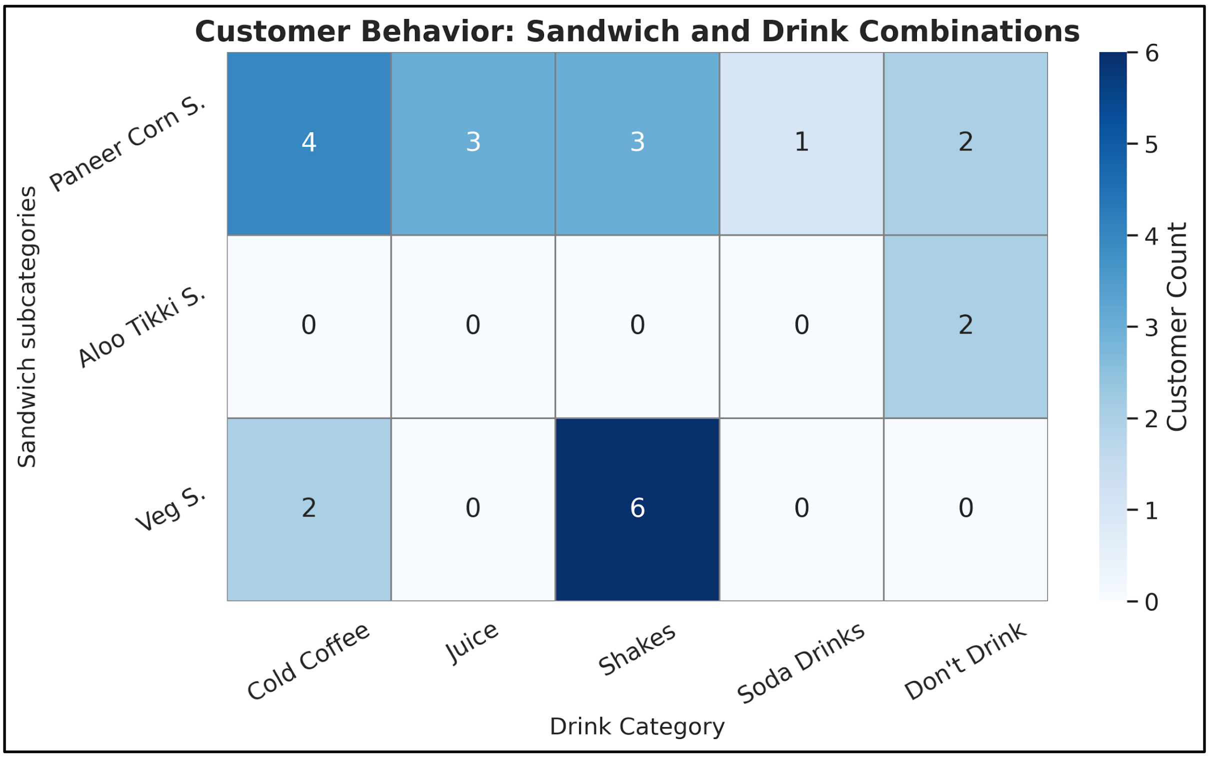
Fig.5 Food and Drink Combination



Food Category	Soda Drinks	Shakes	Juice	Cold Coffee	I don't drink
Grilled Burger	0	2	1	0	0
Patties	1	2	0	1	0
Sandwich	1	9	2	6	5
I don't have food	0	1	4	1	2

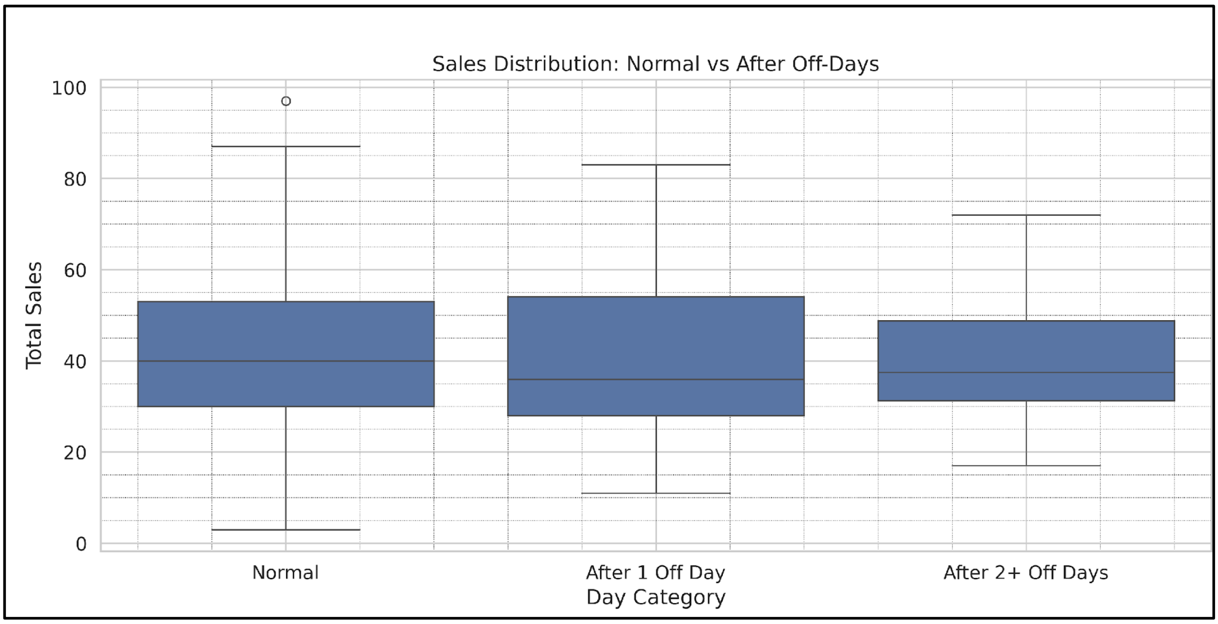
The results presented in **Fig.5** reveal distinct preferences in customer behavior regarding food and drink combinations. Sandwiches emerge as the most favored food option, particularly when paired with shakes, which constitutes the highest customer preference. Additionally, sandwiches consumed with cold coffee or without any beverage also exhibit significant popularity. Conversely, grilled burgers and patties are selected less frequently, and when chosen, they are typically accompanied by shakes. Notably, some customers opt for beverages alone, with juice being the most preferred choice in this category. These findings suggest that customers often engage in specific food and drink pairings, with sandwiches and shakes being particularly prevalent. The moderate variety observed across different food and drink categories indicates potential opportunities for targeted menu promotions, emphasizing popular combinations and catering to customers who prefer beverages independently.

Fig.6 Sandwich Types with Drinks Types Combination



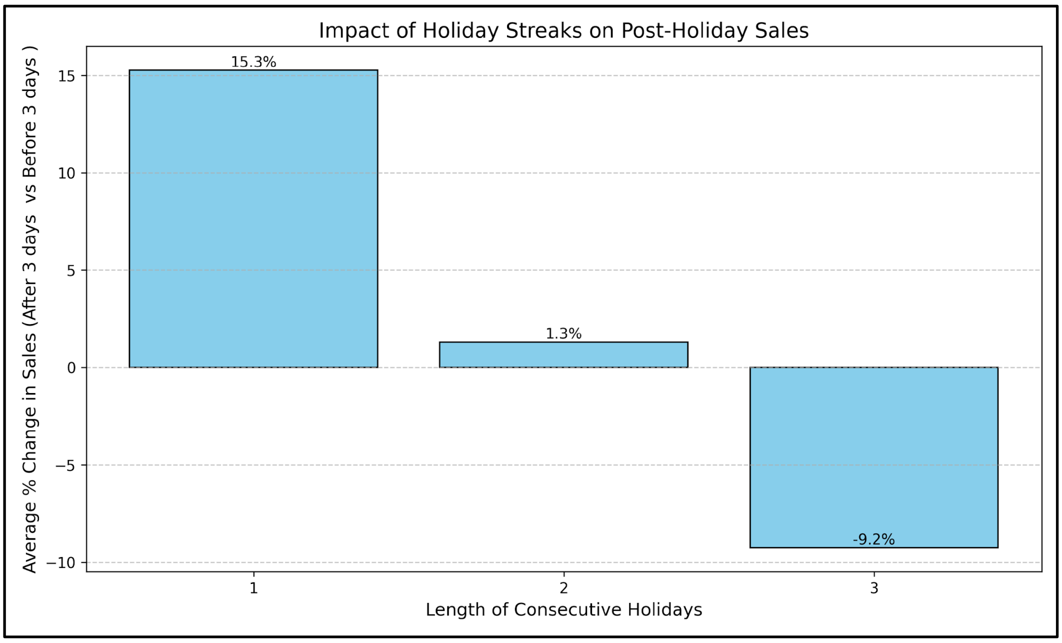
The above heat map graph **(Fig.6)** shows that customers have clear favorite sandwich and drink combinations. The Paneer Corn Sandwich is often paired with cold coffee, juice, and shakes, with cold coffee being the favorite. People who choose the Veg Sandwich mostly prefer shakes, making this the most popular combination overall. There is little interest in soda or not having a drink with the Veg Sandwich. The Aloo Tikki Sandwich is the least chosen and is usually eaten without a drink or sometimes with juice. These results suggest that offering deals or promotions with shakes for Veg and Paneer Corn Sandwiches, or adding more cold drink options for Paneer Corn Sandwiches, could increase sales by matching customer preferences.

Fig.7 Box Plot for sales average sales after Off day until next off day



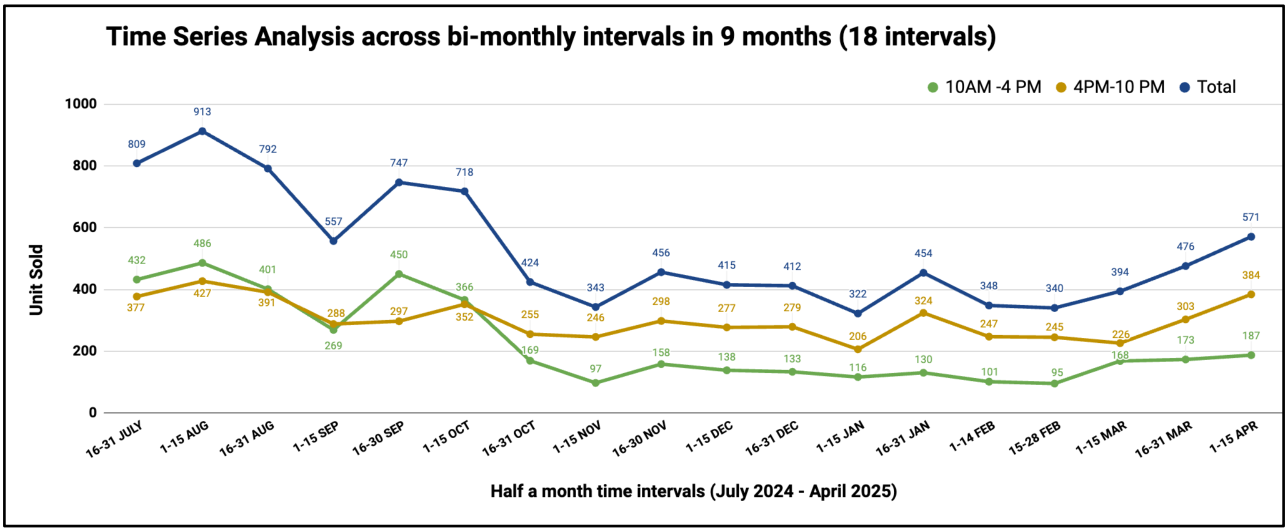
The sales distribution after closures shows a clear pattern in customer behavior. The box plot **(Fig.7)** shows total daily sales divided into 'Normal' days (no previous off-days), 'After 1 Off Day', and 'After 2+ Off Days'. There is a noticeable decrease in the median and overall sales distribution as the number of consecutive off-days increases. 'Normal' days have the highest median sales and the broadest range, showing consistently higher sales. After one off-day, median sales and the overall sales range drop. The biggest effect is seen after two or more consecutive off-days, where median sales fall further, and the sales distribution is the lowest among the three groups. This indicates that longer closures lead to a more significant drop in sales when reopening, suggesting a change in customer buying behavior or a delay in customers returning after longer breaks. This finding highlights the need to consider closure duration when predicting sales and planning for recovery after closures.

Fig.8 Impact of Holiday Streaks until next holiday



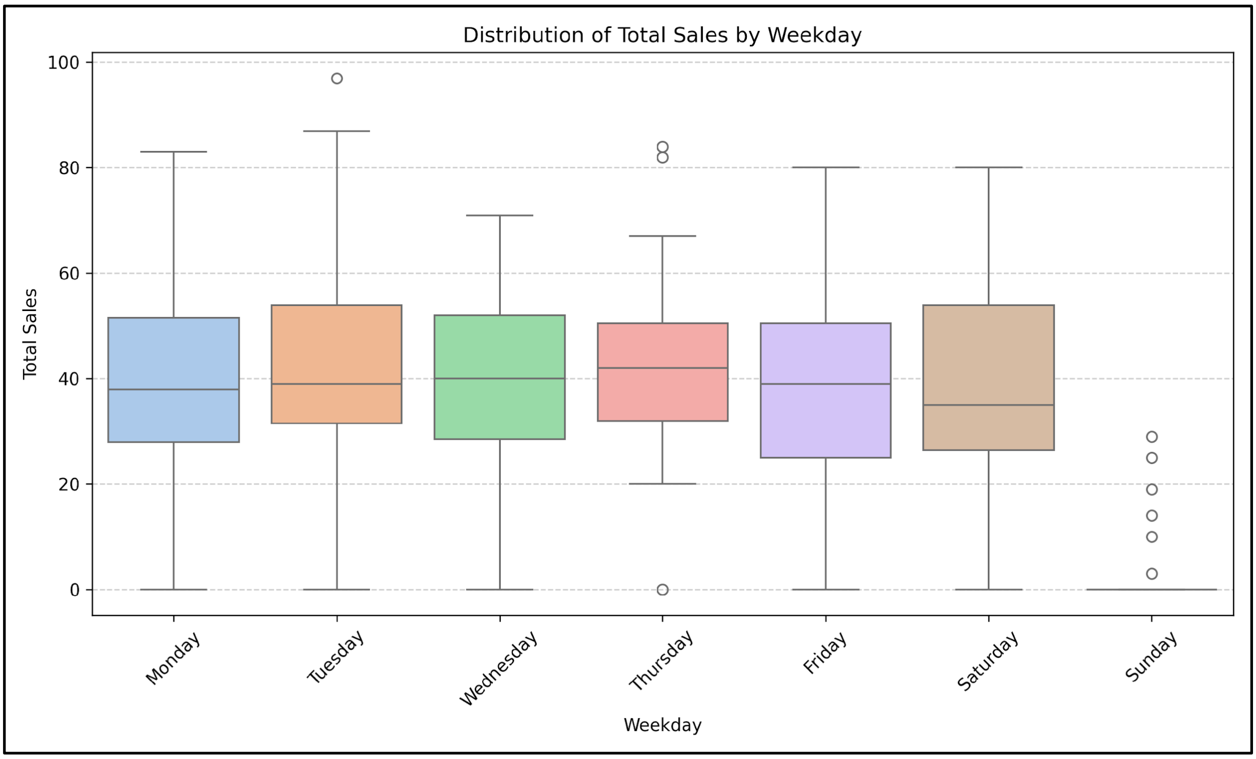
The analysis of sales behavior in relation to holiday streaks reveals a distinct pattern in how consecutive shop closures affect subsequent sales performance. As shown in **Fig.8,** a single-day holiday is associated with a significant positive impact on sales, with an average increase of 15.3% in the three days following the holiday compared to the three days preceding it. This suggests that short breaks may create temporary pent-up demand, leading to a surge in customer purchases once the store reopens. In contrast, two consecutive holidays exhibit a marginal increase in post-holiday sales (1.3%), indicating that the positive momentum tends to diminish with extended closures. Most notably, a streak of three consecutive holidays results in a 9.2% decline in average sales following the closure period, suggesting a possible disruption in customer routines or a loss of momentum in demand. These findings imply that while limited closures can stimulate sales upon reopening, longer streaks may instead dampen sales recovery, potentially due to shifts in consumer buying patterns or deferred demand being met elsewhere.

Fig.9 Time series units sales analysis using line chart



The **Fig.9** shows how many units were sold in half-month intervals over nine months, separated by the time of day: 10AM–4PM and 4PM–10PM. Overall, more units were sold during the 4PM–10PM period than in the earlier part of the day for almost every interval. There are some clear ups and downs, with total sales highest at the beginning of the period (July and August), then dropping through late fall and winter, and finally rising again towards the end (March and April). The biggest drops in sales happened around October to January, especially during 10AM–4PM, when sales were much lower compared to other times. By the last few intervals, sales started to go up again, with both time slots seeing increases but evening sales still higher. This suggests that most customers prefer shopping in the later part of the day, and there are certain times in the year when sales naturally dip or rise, which can help with planning for busy or slow periods.

Fig.10 Distribution of Total Sales by weekday



Looking at **Fig.10**, which shows total sales by weekday, there are important points for deciding when to buy stock for the shop. Sunday has almost no sales, meaning the shop is likely closed or very slow, so it's not smart to buy or restock items as it would slow down cash flow recovery. From Monday to Saturday, sales are higher, with Tuesday, Wednesday, Friday, and Saturday having the most sales, showing faster stock turnover and better cash flow recovery. Monday and Tuesday start the week strong, with Tuesday having some of the highest sales, including occasional very high sales. This makes Monday or Tuesday the best days to buy and restock goods, as items bought at the start of the week will likely sell quickly, improving cash flow recovery. Thursday has slightly lower sales with a more even distribution, meaning sales are steady but not outstanding; restocking is okay but major purchases might not be ideal if cash flow is a worry. Saturday also has strong sales, making it another good day to restock for weekend shoppers, but be careful not to overstock before Sunday when the shop is closed.

3.3 Analyzing Goods Costs and Other Fixed and Variable Costs for a Retail Establishment

In **Fig11**, the analysis of costs for Baked Patty, Sandwich, and Sweet Corns over bi-monthly intervals reveals clear trends in sales activity and resource allocation. Both Baked Patty and Sandwich demonstrate consistently higher costs in most intervals, which, given constant per-item prices, reflects strong and regular sales. In contrast, Sweet Corns incur minimal and often zero costs throughout the observed period, indicating sporadic or limited demand.

Fig.11 Food Cost in Bi-monthly intervals

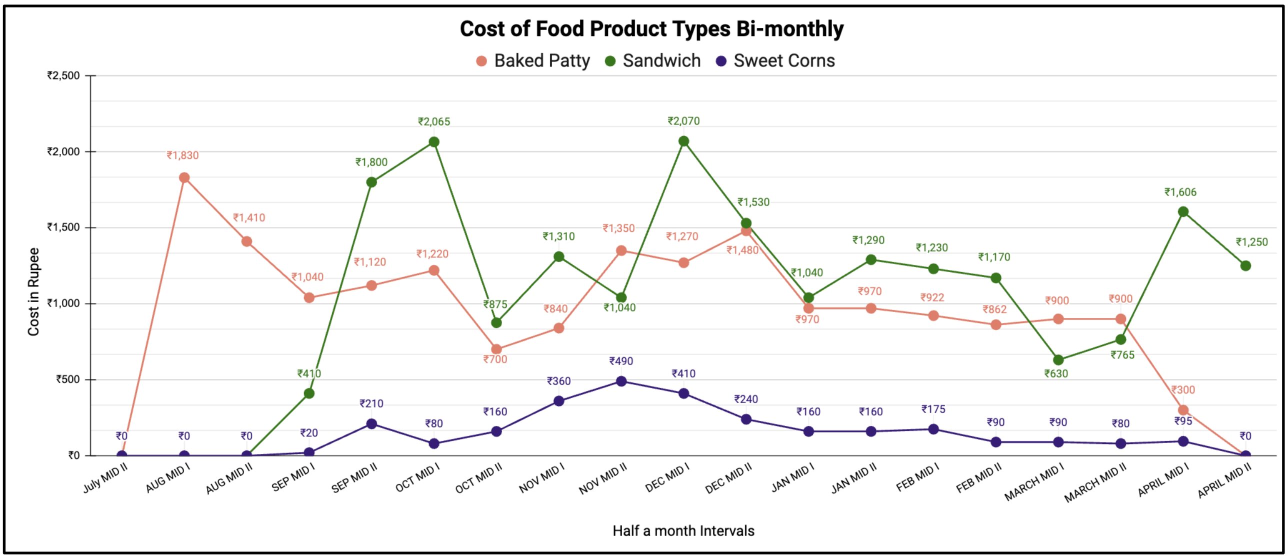
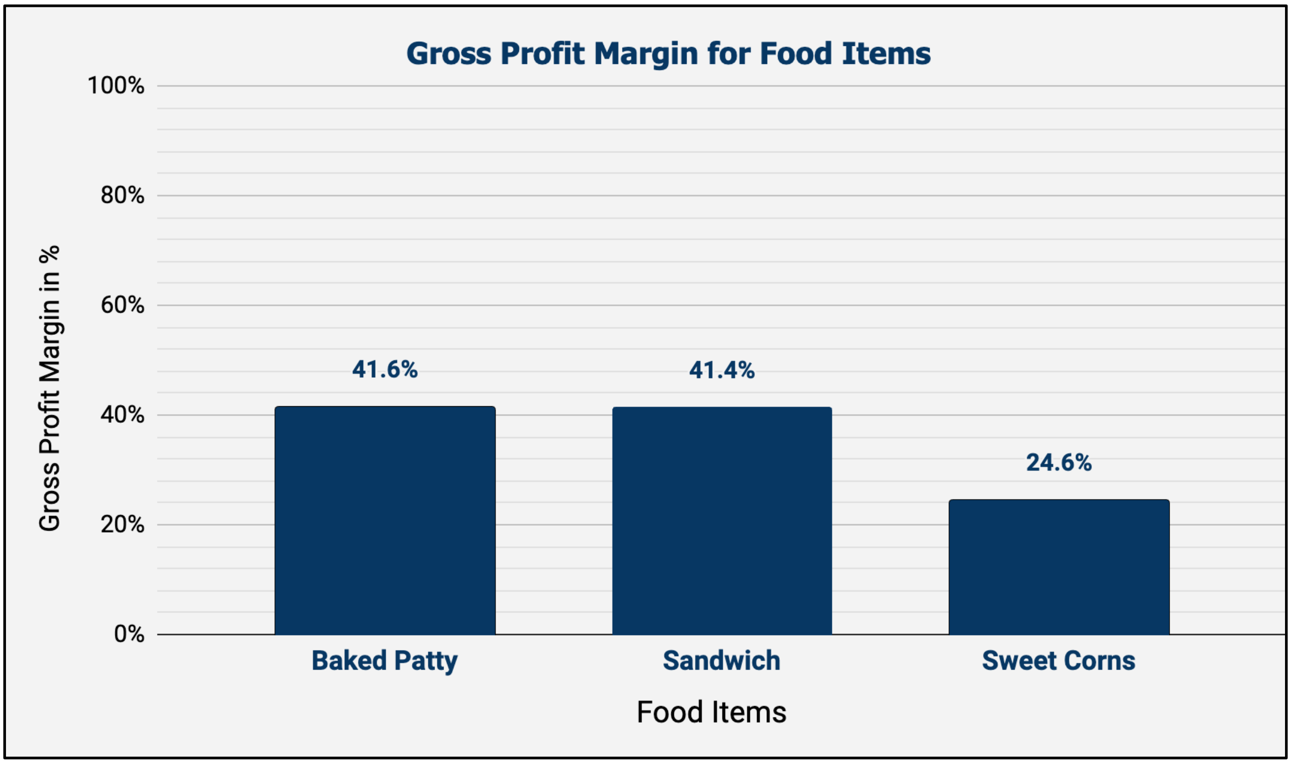
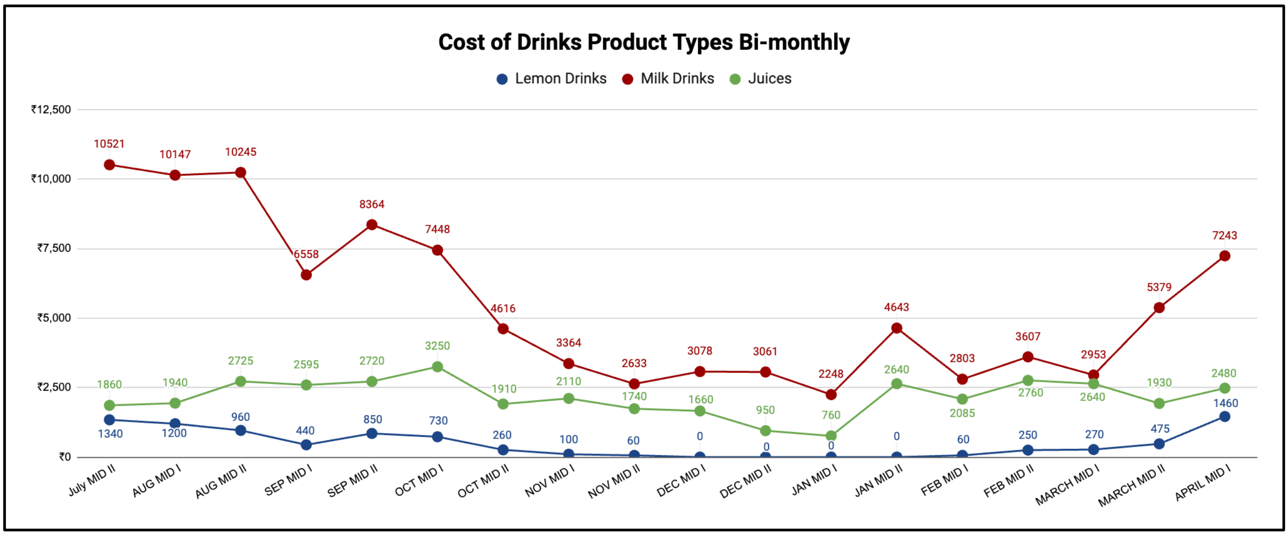


Fig.12 GPM for Food Products



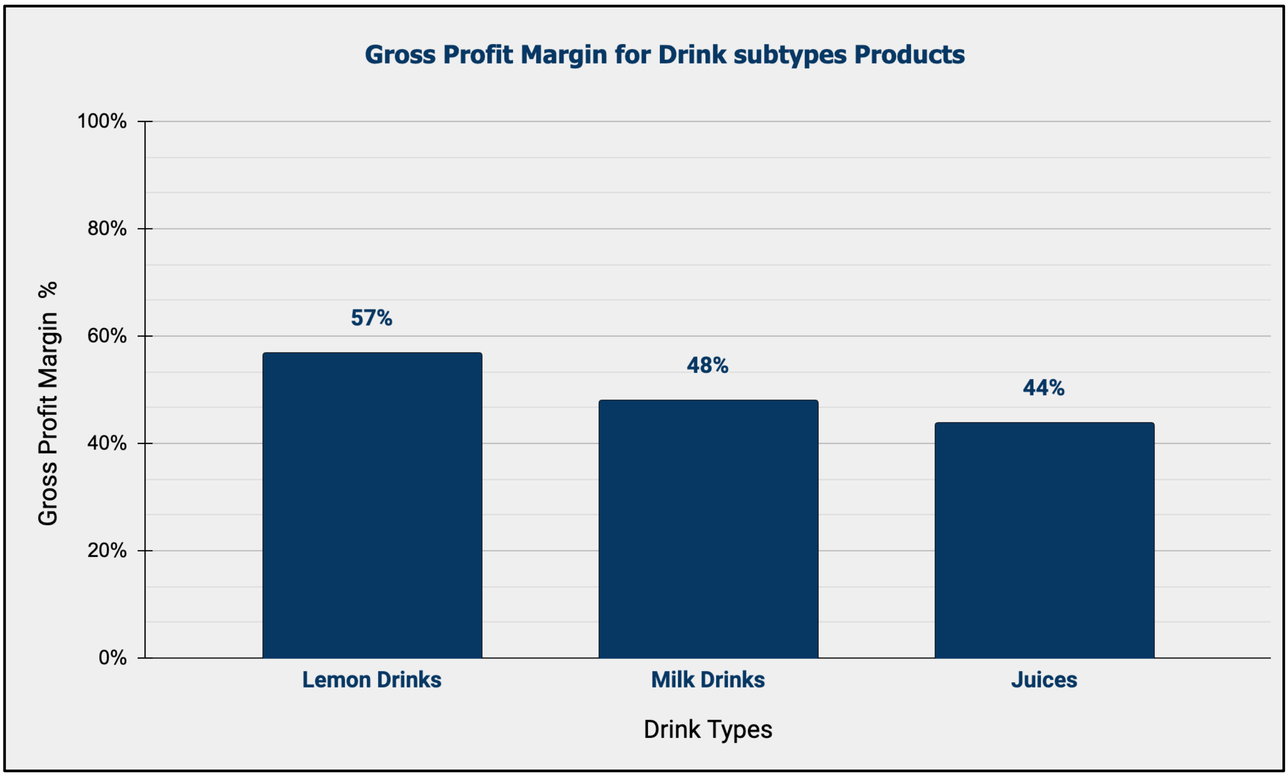
Additionally **Fig.12** shows that gross profit margins are highest for Baked Patty (71.1%) and Sandwich (70.7%), while Sweet Corns lag significantly behind at 32.7%. These patterns suggest that Baked Patty and Sandwich are the primary contributors to profitability and sales volume, whereas Sweet Corns contribute less and have inconsistent sales performance. Overall, the cost and margin data emphasize the value of focusing on high-margin, high-volume items to enhance profitability and resource efficiency. A key fact from the analysis is that Sweet Corns are not only sold independently but are also used as an ingredient in the Sandwich subtype (such as Paneer Sweet Corn Sandwich). This means the gross profit margin for Sweet Corns, as shown in the first chart, may not accurately reflect their true contribution to overall profitability since their use in sandwiches is not separately accounted for in the margin calculation. As a result, Sweet Corns indirectly help generate additional revenue and profit through sandwich sales. This highlights that the Sandwich category supports cross-product value creation by incorporating ingredients like Sweet Corns, allowing these items to contribute to the business both as standalone products and as value-added components in other popular offerings.

Fig.13 Drink Products Cost in Bi-monthly intervals



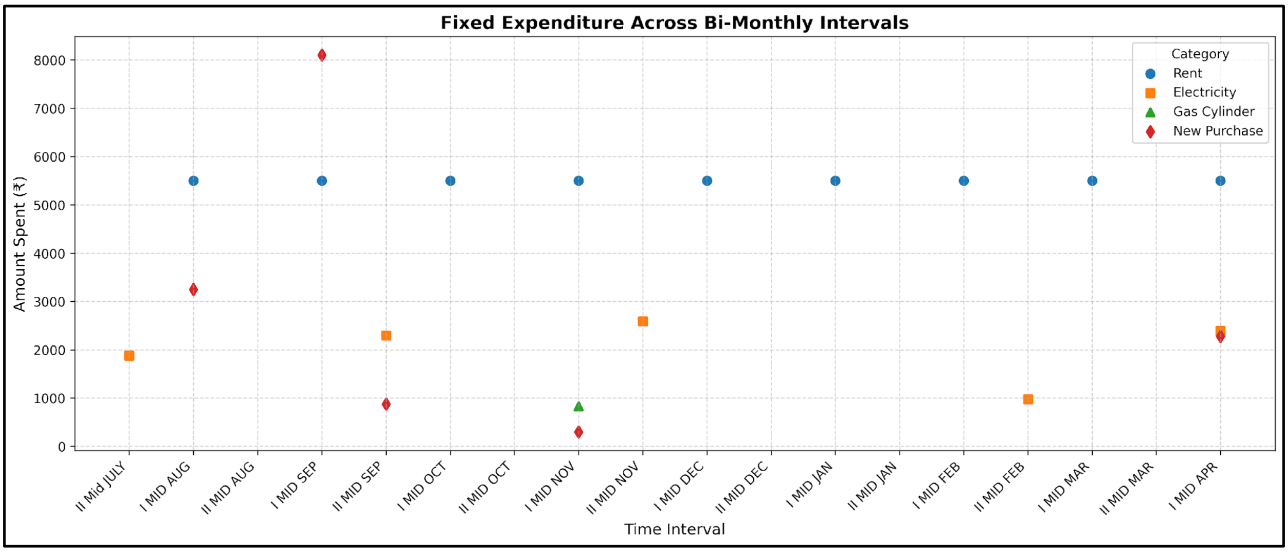
In **Fig13**, the bi-monthly cost chart highlights that Milk Drinks account for the highest and most variable costs, with values regularly above ₹6,000 and peaking at over ₹10,000 in several intervals, indicating Milk Drinks are produced or sold in the greatest volumes. Juices incur moderate and relatively steady costs, fluctuating between ₹1,660 and ₹3,250, reflecting a stable demand over time. Lemon Drinks consistently have the lowest costs, sometimes dropping to zero, which may indicate periods of low sales or seasonal variation, but generally staying below ₹1,500 per interval. These cost patterns, alongside the profit margin data, emphasize that while Lemon Drinks are highly profitable per unit, Milk Drinks contribute most to overall sales volume and cost, and Juices serve as a steady middle ground for both profitability and scale.

Fig.14 GPM for Drink Products



The gross profit margin chart for drink subtypes reveals that Lemon Drinks have the highest margin at 132%, indicating exceptionally strong profitability per sale. Milk Drinks also perform well, with a gross profit margin of 93%, while Juices show a still-healthy margin at 78%. This ranking suggest that all drink types are profitable, but selling Lemon Drinks yields the greatest return above cost.

Fig.15 Fixed Cost in Bi-monthly intervals



As observed in **Fig.15, Rent is** a consistent, fixed expense occurring every other interval with a value of ₹5500, represented by purple circular markers. **Electricity** bills are less frequent and show variability in amount, represented by brown square markers at irregular intervals with amounts ranging from ₹1000 to ₹2600. **Gas** cylinder purchases are infrequent and appear as single events marked by pink triangular markers, with only one recorded at around the "II MID NOV" interval with a value of ₹832. **New item purchases** are also infrequent and show the highest variability in expenditure, ranging from ₹300 to over ₹8000, represented by teal diamond markers appearing sporadically. The plot clearly distinguishes between recurring fixed costs like rent and more sporadic fixed costs like electricity bills, gas cylinder purchases, and new item acquisitions, with their frequency and magnitude visually apparent.

Fig.16 Variable Cost in Bi-monthly intervals

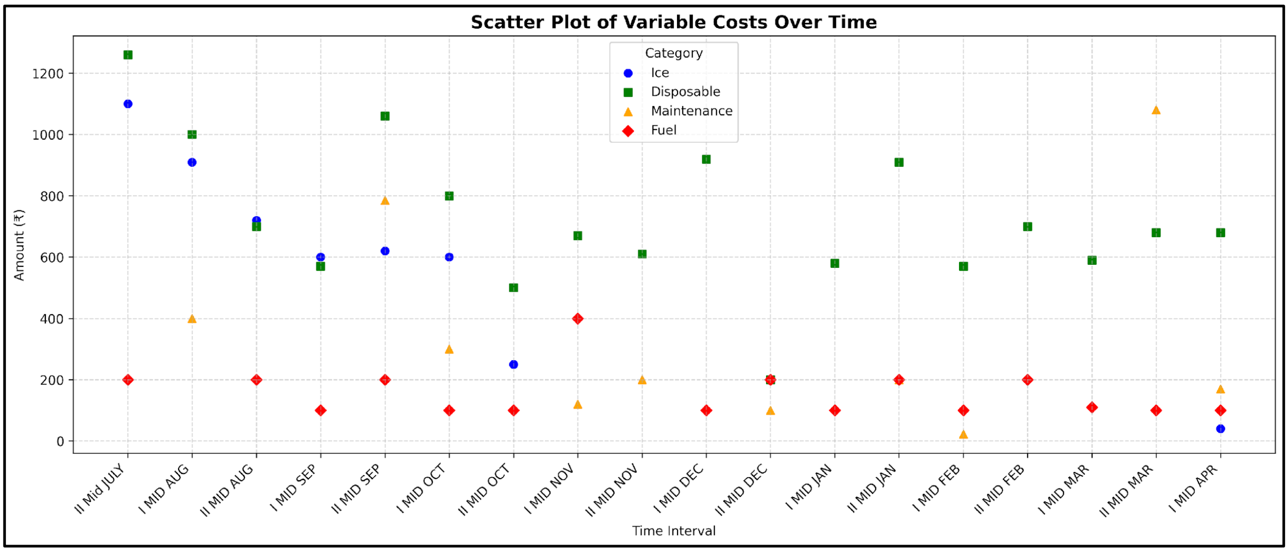


Fig.16 indicates that disposable costs consistently remain the highest variable expense, exerting significant financial pressure on the shopkeeper. This suggests that exploring alternatives such as reusable items or bulk purchasing could effectively reduce these costs. **Ice** costs exhibit a clear seasonal pattern, peaking during the warmer months of July to October and decreasing in colder periods, which underscores the need for efficient storage and management to minimize waste during high-demand times**. Maintenance** costs, although sporadic, can be substantial when they occur, highlighting the importance of proactive maintenance to mitigate these unexpected expenses. **In contrast**, fuel costs are relatively low and consistent, presenting a smaller financial burden. The findings suggest that disposables and maintenance are primary areas for cost reduction, with potential savings achievable through sustainable alternatives and optimized maintenance practices. Additionally, while fuel costs are lower, optimizing delivery routes or adopting energy-efficient practices could further contribute to overall cost savings.

Section 4. Interpretation of Results and Recommendations

4.1 Interpretation of Results

The analysis in Section 3.1 provides a comprehensive overview of revenue and product demand across 18 half-month intervals over nine months. A significant insight is the Pareto effect in revenue distribution, where 38% of high-performing SKUs consistently account for 80% of total revenue. This trend is mirrored in units sold, where the top 35% of SKUs contribute to 80% of volume sales, indicating strong customer preferences for certain fast-moving items.
External influences on revenue show that the number of unique products sold correlates more strongly with revenue than the number of working days. For instance, periods with fewer working days experienced higher sales due to a diverse product assortment, suggesting assortment diversity as a key sales driver.

Section 3.2 explores pricing and customer behavior. Products priced at ₹40 exhibit the highest normalized unit sales and variety, indicating a psychological price-value sweet spot. Food-drink combinations show a preference for sandwiches paired with cold coffee or shakes, suggesting that bundled promotions could enhance sales.
Operational closures significantly impact sales, with median sales declining sharply after two or more off-days. A three-day holiday streak leads to a 9.2% post-closure sales drop, while a one-day holiday results in a 15.3% increase, indicating demand surges for short closures and disrupted consumer patterns for longer ones. Time-of-day sales analysis shows 4PM–10PM slots consistently outperform earlier hours, with seasonal dips in October to January and rebounds in March-April, aiding capacity planning. Weekday analysis reveals negligible sales on Sundays, unsuitable for restocking, while strong turnover on Mondays and Tuesdays suggests optimal days for inventory replenishment.

Section 3.3 shows that Baked Patty and Sandwich categories have the highest and most consistent costs. This means they sell more and make more money, with gross margins over 70%. Sweet Corns add value to sandwiches, so they contribute to sandwich and will be considered as good product. Beverages, with Milk Drinks having the highest costs and sales volume, and Lemon Drinks having the highest gross margins, show that they want to sell a lot and make a lot of money. Rent is a stable expense, but disposables are the biggest cost, with ice and maintenance having seasonal and occasional patterns. To save money, we should focus on disposables and maintenance. We should keep our money invested in high-margin, high-demand items so we can make more money.

4.2 Recommendations:

1. Milk Soda, Mix Juice, Hot Coffee (Med), Aloo Patties, Banana Shake, Mango Shake, Mousambi Juice, Paneer Corn Sandwich, Veg Sandwich, and Lemon Soda are key contributors to revenue and sales. Retain them as core offerings and expand variety in these. (Fig 1 & 2)
2. Introduce low-selling, high-revenue SKUs in combo meals with high-selling items to boost volume. (Fig 1 & 2)
3. Focusing on increasing the variety of products sold has a greater positive impact on revenue than simply increasing the number of working days within an interval. (Fig 3)
4. There are 15 number of SKU available in the 40 rupees bin and they contribute high unit sales after 30 rupee bin, prices of some needful or most likely products in this range can be increase by 10 rupees to get the more profit. (Fig.4)
5. Introduce Sandwich–Shake Combos: Launch combo offers around Paneer Corn and Veg Sandwiches with popular shakes (Banana, Oreo, Custom) to leverage existing demand. (Fig.6)
6. Promote Underperforming Items: Explore why items like Aloo Tikki Sandwich and repositioning, or recipe changes. (Fig.6)
7. I suggest the shop stay open for no more than one extra day during the holiday season. This will keep sales momentum going and prevent customer dissatisfaction from extended closures. (Fig. 7 & 8)
8. If urgent work requires a day off, consider taking the morning off but keep the shop open in the evening, as this period is crucial for maintaining steady sales and supporting business recovery. Focus your efforts on maximizing evening attendance rather than depending on the slower morning slot. (Fig.9)
9. Purchase most stock and fresh goods on Monday or Tuesday, taking advantage of higher sales turnover through the middle of the week to recover cash flow rapidly. (Fig.10)
10. Apply offers on item on Saturday where there is less sales and to have good cash flow for purchasing goods on Monday or Tuesday and release previous stock. (Fig.10)
11. Utilize digital networks for advertising and reach out to newspapers to publish his weekly or monthly offers.
12. Target of the each interval should be more than 30,000 to recover the loss and then have a enough money to at least cover fixed investment which more than 6000 per month. (Fig.15)
13. Utilize digital networks for advertising and reach out to newspapers to publish his weekly or monthly offers with ice-cream to replace ice cost. (Fig.16)

4.3 Conclusion

By implementing these recommendations, shop owner can grow his business to incoporate major problems like less sales and organize menu items and improve menu items and manage his inventory and strategically spend on new purchases.

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************ End of Report ************